AI 101 for Mechanical Engineers · Module 5

Scaling and encoding

Lesson 5.2

What you will learn

> Calculate min-max scaling and standardization, distinguish transformed values from original measurements, and represent categories without inventing numerical order.

Lesson overview

Module 5 · Lesson 2

Scaling and encoding

AI 101 for Mechanical Engineers

1 Min-max scaling

$$x_{\text{scaled}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

x: the value · $x_{\min}$, $x_{\max}$: the stated bounds, in the same unit as x

Example: x = 25 °C; bounds 0–100 °C $\quad x_{\text{scaled}} = \frac{25 - 0}{100 - 0} = 0.25$

The stated bounds map to 0 and 1. Later values may fall outside that range.

2 Standardization

$$z = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}}$$

μ_{train}: training mean

σ_{train}: training standard deviation

Example: x = 60 °C; μ_{train} = 40 °C; σ_{train} = 10 °C $\quad z = \frac{60 - 40}{10} = 2$

3 Categories do not need a false order

Coding A, B and C as 1, 2 and 3 can imply an unsupported order and distance.

One-hot encoding uses separate indicator columns:

Category	Is A	Is B	Is C
A	1	0	0
B	0	1	0
C	0	0	1

1 means membership; 0 means no membership. The indicators are dimensionless.

4 Fit once, apply consistently

Training data → Fit and store rules → Apply stored rules to validation and test inputs

- Do not refit on validation or test data.
- Declare a rule for zero denominators and for unseen categories.
- Scaling matters most for distance-based and neural methods, not tree splits.

Key takeaway

Choose the transformation for the method and data. Keep training statistics fixed.

Transformed values are dimensionless. Keep the original measurements and units.

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State the scaling rule exactly

- Distance-based methods and penalized regression are sensitive to feature scale [16]
- Neural-network training is commonly done on scaled inputs
- Scaling transforms magnitudes using a stated mathematical rule
- Avoid ambiguous terms such as “normalization”

Min-max scaling maps stated bounds to 0 and 1

$$x_scaled = (x - x_{min}) \,/\, (x_{max} - x_{min})$$

$$x_scaled = 25 \,/\, 100 = 0.25$$

- Min-max scaling uses stated bounds x_{min} and x_{max} [33]
- x, x_{min} and x_{max} share a unit; x_scaled is dimensionless
- Example: bounds 0–100 °C and x = 25 °C

Standardization uses the training mean and standard deviation

$$z = (x - \mu_{train}) / \sigma_{train}$$

$$z = (60 - 40) / 10 = 2$$

- μ_{train} and σ_{train} come from training data [16]
- Example: 60 °C, training mean 40 °C, standard deviation 10 °C
- z is dimensionless; 25 °C with same statistics gives −1.5

Keep originals; reuse stored training statistics

- Keep original measurements and units for physical interpretation
- A dimensionless value does not replace that record
- Apply stored training statistics to validation and test inputs
- A constant feature gives a zero denominator: declare a rule

Encode categories without inventing order

Integer codes

- Types A, B and C become 1, 2, 3
- Can impose unsupported order and distance

One-hot encoding

- Separate binary indicator columns [34]
- Show membership, not physical magnitude

Fit category handling on training data and declare how an unseen category is treated.

One-hot indicators for three known categories

Category	Indicator columns
A	[1, 0, 0]
B	[0, 1, 0]
C	[0, 0, 1]

> Indicators are dimensionless. Do not invent an order just because a model needs numbers.

Choose scaling for the method and data

- Matters most for distances or magnitude-sensitive optimization
- Tree splits generally unchanged by consistent monotonic rescaling
- Min-max does not keep future values inside the range
- Standardization is not a general outlier remedy

Tool names are not a reason to transform

- StandardScaler and OneHotEncoder are common scikit-learn implementations
- See the official documentation for your version [91]
- Fit on training data only, then apply consistently
- Check the effect on validation data

CHECK YOUR UNDERSTANDING

Standardize 60 °C using a training mean of 40 °C and standard deviation of 10 °C.

Answer before moving on.

CHECK YOUR UNDERSTANDING

Standardize 60 °C using a training mean of 40 °C and standard deviation of 10 °C.

> z = (60 − 40) / 10 = 2, a dimensionless value. Reuse the training statistics when transforming test inputs.

Key takeaways

- State the exact scaling equation
- Retain original units; transformed outputs are dimensionless
- Apply fitted training statistics consistently
- Declare rules for constant features and unseen categories
- One-hot indicators avoid unsupported numerical ordering

References

[16] G. James, D. Witten, T. Hastie, and R. Tibshirani, An Introduction to Statistical Learning: With Applications in R, 2nd ed. New York, NY, USA: Springer, 2021. doi: 10.1007/978-1-0716-1418-1

[33] J. Han, M. Kamber, and J. Pei, Data Mining: Concepts and Techniques, 3rd ed. Waltham, MA, USA: Morgan Kaufmann, 2012. doi: 10.1016/C2009-0-61819-5

[34] F. Pedregosa et al., “Scikit-learn: Machine learning in Python,” J. Mach. Learn. Res., vol. 12, pp. 2825–2830, 2011.

[91] scikit-learn developers, scikit-learn User Guide and API Reference, ver. 1.9.1. [Online]. Available: https://scikit-learn.org/stable/ Accessed: Oct. 20, 2026.

Numbers match the course reference list. The written lesson is the authoritative text.